

## 7.5 Electrical Characteristics (continued)

Typical conditions are measured at 25°C with nominal BAT voltages of 18 V (BQ76920), 36 V (BQ76930), or 48 V (BQ76940) with  $V_{CELL} = 4$  V. Min and max values include full recommended operating condition temperature range from -40°C to +85°C. Certain characteristics may be shown at different voltage or temperature ranges, as clarified in the Test Condition sections.

| PARAMETER                              |                                | TEST CONDITION                                             | MIN    | TYP    | MAX    | UNIT     |
|----------------------------------------|--------------------------------|------------------------------------------------------------|--------|--------|--------|----------|
| CC <sub>GAIN</sub>                     | Gain error                     | Over input voltage range                                   |        | ± 0.5% | ± 1.5% | FSR      |
| CC <sub>GAINDRIFT</sub>                | Gain error drift               | Over input voltage range                                   |        |        | 150    | PPM / °C |
| CC <sub>RIN</sub>                      | Effective input resistance     |                                                            |        | 2.5    |        | MΩ       |
| <b>THERMISTOR BIAS</b>                 |                                |                                                            |        |        |        |          |
| R <sub>TS</sub>                        | Pull-up resistance             | T <sub>A</sub> = 25°C                                      | 9.85   | 10     | 10.15  | kΩ       |
| R <sub>TSDRIFT</sub>                   | Pull-up resistance across temp | T <sub>A</sub> = -40°C to 85°C                             | 9.7    |        | 10.3   | kΩ       |
| <b>DIETEMP</b>                         |                                |                                                            |        |        |        |          |
| V <sub>DIETEMP25</sub>                 | Die temperature voltage        | T <sub>A</sub> = 25°C                                      |        | 1.20   |        | V        |
| V <sub>DIETEMPDRIFT</sub>              | Die temperature voltage drift  |                                                            |        | -4.2   |        | mV/°C    |
| <b>INTEGRATED HARDWARE PROTECTIONS</b> |                                |                                                            |        |        |        |          |
| OV <sub>RANGE</sub>                    | OV threshold range             |                                                            | 0x2008 |        | 0x2FF8 | ADC      |
| UV <sub>RANGE</sub>                    | UV threshold range             |                                                            | 0x1000 |        | 0x1FF0 | ADC      |
| OV <sub>UVSTEP</sub>                   | OV and UV threshold step size  |                                                            |        | 16     |        | LSB      |
| UV <sub>MINQUAL</sub>                  | UV minimum value to qualify    | Below UV <sub>MINQUAL</sub> , the cell is shorted (unused) |        | 0x0518 |        | ADC      |
| OV <sub>DELAY</sub>                    | OV delay timer options         | OV delay = 1 s                                             | 0.7    | 1      | 1.75   | s        |
|                                        |                                | OV delay = 2 s                                             | 1.6    | 2      | 2.75   |          |
|                                        |                                | OV delay = 4 s                                             | 3.5    | 4      | 5      |          |
|                                        |                                | OV delay = 8 s                                             | 7      | 8      | 10     |          |
| UV <sub>DELAY</sub>                    | UV delay timer options         | UV delay = 1 s                                             | 0.7    | 1      | 1.75   |          |
|                                        |                                | UV delay = 4 s                                             | 3.5    | 4      | 5      |          |
|                                        |                                | UV delay = 8 s                                             | 7      | 8      | 10     |          |
|                                        |                                | UV delay = 16 s                                            | 14     | 16     | 20     |          |
| OCD <sub>RANGE</sub>                   | OCD threshold options          | Measured across (SRP - SRN) <sup>(2)</sup>                 | 8      |        | 100    | mV       |
| OCD <sub>STEP</sub>                    | OCD threshold step size        | RSNS = 0                                                   |        | 2.78   |        | mV       |
|                                        |                                | RSNS = 1                                                   |        | 5.56   |        | mV       |
| OCD <sub>DELAY</sub>                   | OCD delay options              | See Note <sup>(3)</sup>                                    | 8      |        | 1280   | ms       |
| SCD <sub>RANGE</sub>                   | SCD threshold options          | Measured across (SRP - SRN) <sup>(2)</sup>                 | 22     |        | 200    | mV       |
| SCD <sub>STEP</sub>                    | SCD threshold step size        | RSNS = 0                                                   |        | 11.1   |        | mV       |
|                                        |                                | RSNS = 1                                                   |        | 22.2   |        | mV       |
| SCD <sub>DELAY</sub>                   | SCD delay options              |                                                            | 35     | 70     | 105    | μs       |
|                                        |                                |                                                            | 50     | 100    | 150    | μs       |
|                                        |                                |                                                            | 140    | 200    | 260    | μs       |
|                                        |                                |                                                            | 280    | 400    | 520    | μs       |
| t <sub>PROTACC</sub>                   | Delay accuracy for OCD         |                                                            | -20%   |        | 20%    |          |
| OC <sub>OFFSET</sub>                   | OCD and SCD voltage offset     |                                                            | -2.5   |        | 2.5    | mV       |
| OC <sub>SCALEERR</sub>                 | OCD and SCD scale accuracy     |                                                            | -10%   |        | 10%    |          |
| <b>CHARGE AND DISCHARGE DRIVERS</b>    |                                |                                                            |        |        |        |          |